Disha: Vaastu Scanner App

Concept and technical design for a sensor-based Vaastu home analysis tool

The concept

Most Vaastu apps are static articles or a plain compass. Disha uses the phone's own sensors to **measure** the home and give room-by-room guidance, with practical fixes that don't require demolition. That makes it useful for renters and apartment dwellers, not just people building a house.

The scanner has four layers: get an accurate North, map rooms to Vaastu zones, apply the rules, and generate remedies and a report.

1. Getting an accurate North

This is the hardest part and where most compass apps fall short.

Sensor fusion

The phone combines the **magnetometer** (Earth's magnetic field), **accelerometer** (gravity, for tilt) and **gyroscope** (smooth rotation) into a stable, tilt-compensated heading. On iOS this comes from Core Motion / Core Location heading; on Android from the rotation vector sensor.

Magnetic to true North

A compass points to magnetic North, but Vaastu uses true (geographic) North. The difference, called **declination**, depends on location. The app takes the GPS latitude/longitude and runs the World Magnetic Model (free libraries exist, e.g. `geomagnetism` for JavaScript) to get the correction. In New Jersey it is roughly 12-13° West. Since each zone is only 22.5° wide, skipping this can put a room in the wrong zone.

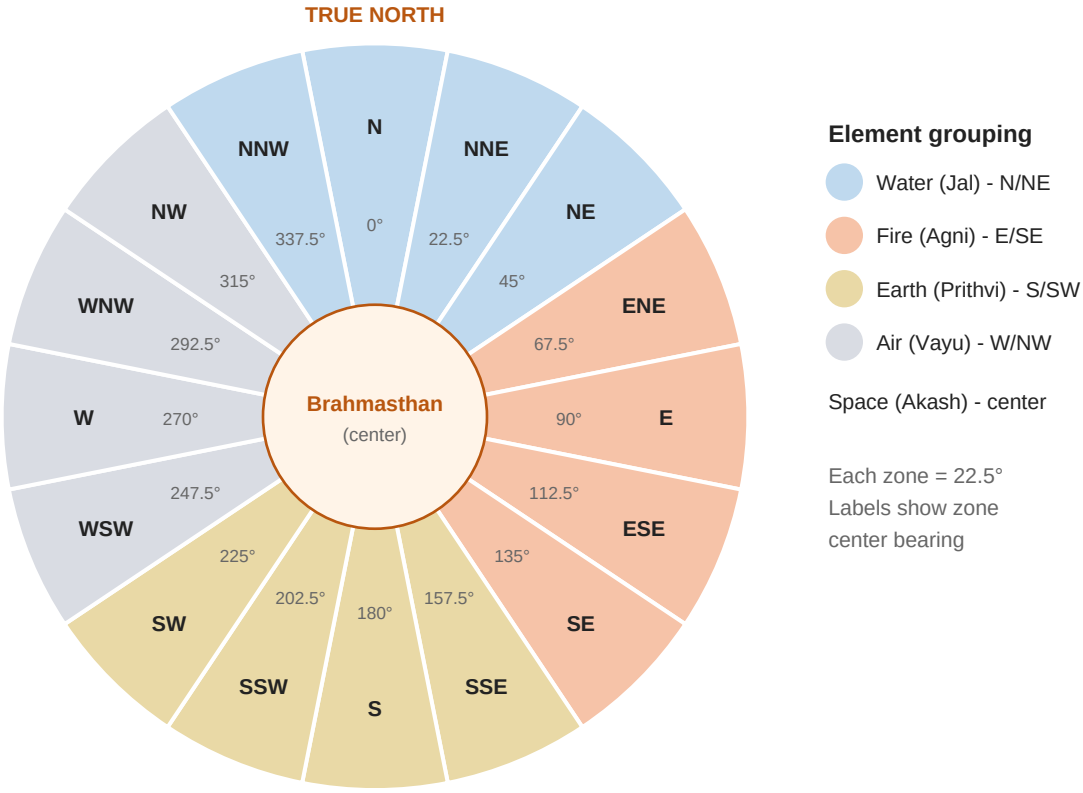
Interference check

Indoors, steel beams, appliances, speakers and rebar distort readings. The app compares the measured field strength with what the World Magnetic Model expects at that location (around 50 µT in the US). If it is far off, the app warns "move away from metal" or "reading unreliable here" and can prompt the figure-8 calibration motion.

Web limitation: a browser exposes only the heading, not raw field strength, so the interference check needs a native build (Xcode / Android Studio). The web version can use iOS's `webkitCompassAccuracy` as a rough quality signal.

2. Mapping rooms to zones

Vaastu divides the home into 16 directional zones of 22.5° each around the center, the Brahmasthan.



Option A: Point-and-tag (no drawing)

- The user stands roughly at the center of the home.
- They point the phone toward each room and tag it: Kitchen, Bedroom, Toilet, Main door.
- The app records the true bearing for each. For example, 135° falls in the Southeast zone, the fire (Agni) zone.

Option B: Floor plan

- The user uploads a photo of the floor plan.
- They point the phone along one wall shown on the plan, so the app learns the plan's rotation relative to North.
- They tap rooms on the plan. The app finds the center, overlays the 16-zone grid and computes each room's zone geometrically.

Option B is more accurate and makes a better visual report; Option A is faster and needs no plan. Offering both is ideal.

3. Rule engine

A lookup table of room type against zone, returning a verdict and the traditional reasoning:

Room	Favorable zones	Avoid
Kitchen	SE, then NW	NE, SW
Master bedroom	SW	NE, SE
Pooja room	NE	S, SW, near toilet
Toilet	WNW, SSW	NE, center
Main door	Specific auspicious padas on each side	Several inauspicious padas

Each room gets a score and the home gets an overall score. Store the rules in a JSON file so they can be refined, or regional variations added, without code changes.

4. Remedies and report

Every problem maps to non-structural fixes:

- Change the bed's head direction or desk facing direction
- Use specific colors for walls or decor
- Rearrange furniture; adjust mirror placement
- Add elements such as plants, water features or metal objects

The output is a home diagram with color-coded zones, a room-by-room breakdown and remedies, exportable as a PDF.

Features summary

Feature	What it does	Tier
True-North compass	Declination-corrected, tilt-compensated heading with accuracy indicator	Free
Desk / bed checker	Point the phone, get an instant verdict on facing direction	Free
Single-room check	Zone and verdict for one room	Free
Full-home scan	All rooms, overall score, remedies	Paid
PDF report	Shareable/printable color-coded report	Paid
Saved properties	Multiple homes; house-hunting mode for viewings	Paid

Accuracy caveats to design around

- Holding the phone flat and level matters; show a bubble-level indicator.
- Steel-frame apartment buildings give poor readings. Let the user take a reading near a window or outside, then align the floor plan to it.
- Vaastu is a traditional practice; frame results as "per traditional Vaastu guidelines" rather than guarantees.